

Application of Superconductivity

Zero resistance. Zero loss. The invisible technology bulding our future.

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Superconductivity

Zero Resistance, Zero Loss

Normal Wires

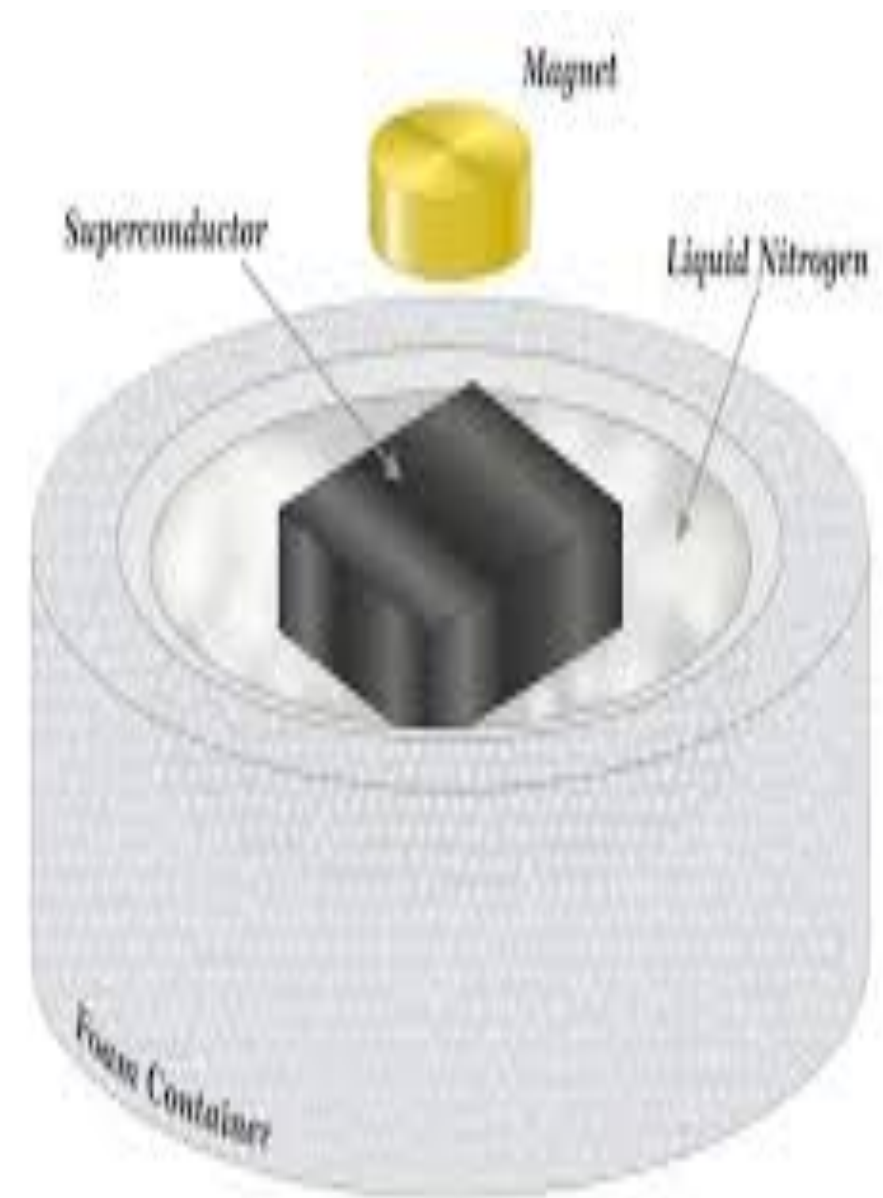
Copper wires get hot and waste
~10% of electricity as heat.

Superconductors

Zero electrical resistance — no
energy lost, no heat generated.

The Catch

They only work below -200°C (-328°F). They also levitate magnets via
the Meissner Effect.





Seeing Inside the Human Body

1. MRI machines use very strong superconducting magnets.
2. They create detailed images of our organs, brain, and joints.
3. Without superconductors, MRI magnets would be too weak and cost too much electricity.

Result: Doctors detect cancer and injuries early — saving lives every day.



APPLICATION 2- Maglev Trains

The Fastest Trains on Earth

Magnetic Levitation

Superconductors make the train float 4 inches above the track — no contact, no friction.

Extreme Speed

Up to **375 mph (600 km/h)** — faster than Formula 1 cars.

Example

Japan's SC Maglev train — silent, smooth, and revolutionary.

Smashing Atoms to Discover the Universe

- 1.The LHC (Large Hadron Collider) at CERN is the world's largest superconducting machine.
- 2.Superconducting magnets bend beams of particles moving near light speed.
- 3.This helped discover the "God Particle" (Higgs Boson).

This led to the discovery of the **Higgs Boson** — the "God Particle."



Fixing the "Wasted Energy" Problem

Today's Problem

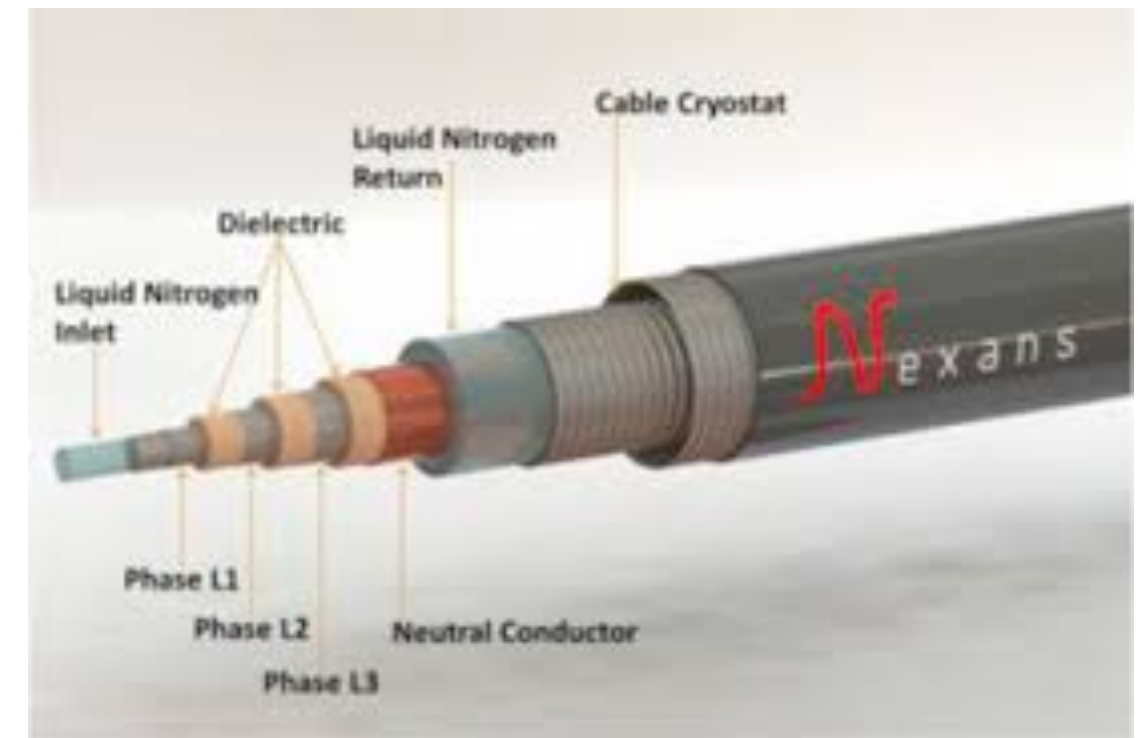
We lose **5–10%** of all electricity as heat in conventional power lines.

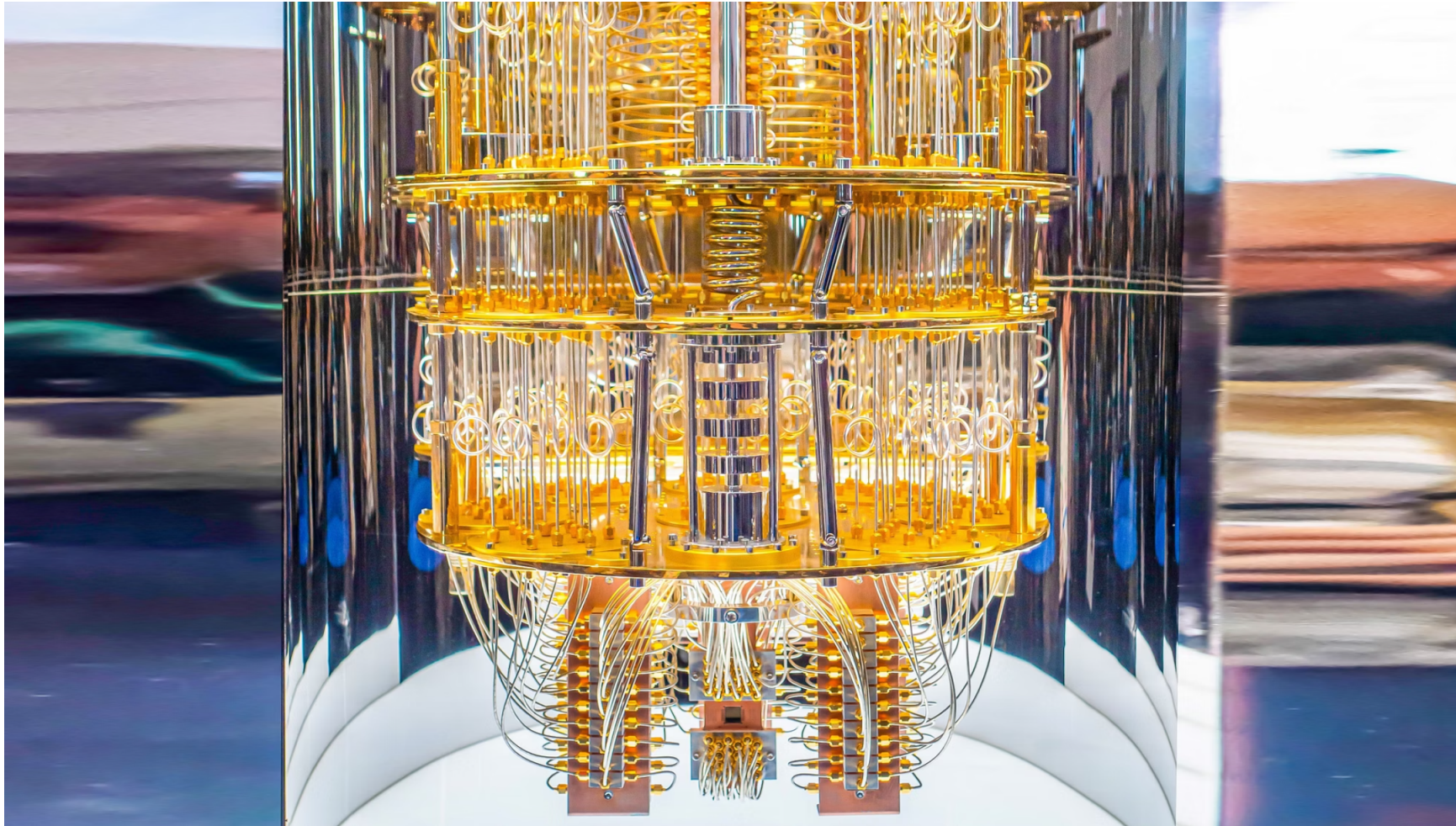
Superconducting Solution

Superconducting cables carry **10x more power** with **0% loss**.

Ideal For

Dense cities like Delhi and Mumbai — more power without digging new tunnels.





The Brains of the Future

1. Google and IBM are building Quantum Computers.
2. The core part (Qubit) is a tiny superconductor.
3. It allows computers to solve problems in seconds that would take normal computers thousands of years.

Superconductors are not just for big magnets — they are the key to quantum computing.



The Problem: Keeping Things Cold

Current Cost

Cooling requires liquid helium or nitrogen — expensive and complex infrastructure.

The Goal

Find a **Room-Temperature Superconductor** and everything changes.

The Potential

Laptops that never overheat, super-efficient flying cars, zero-loss power grids everywhere.

Why It Matters

Medicine

MRI saves lives with superconducting magnets.

Transport

Maglev trains are the future of speed.

Energy

Zero-loss power lines for dense cities.

Computing

Quantum computers powered by superconducting qubits.



"Superconductivity is not just physics — it is the invisible technology building our future."